

SUBJECT OUTLINE**Subject code and name**

DLE602 Deep Learning

SECTION 1 – GENERAL INFORMATION**1.1 Administrative details**

Associated award/s	Duration	Level
Master of Software Engineering (Artificial Intelligence) Master of Software Engineering (Artificial Intelligence, Advanced)	1 Trimester (12 weeks, or intensive over 6 weeks)	600

1.2 Core or elective subject

- ☐ core subject (within associated awards)
- ☐ elective subject (within associated award/s, and in other approved awards)
- ☒ other (please specify below):

Specialisation

1.3 Subject weighting

Subject credit points	Total course credit points
10	Master: 120 credit points Master (Advanced): 160 credit points

1.4 Student workload

Duration	Facilitated study h/p/w	Personal study h/p/w	Total study h/p/w
12-week duration	3	7	10
6-week duration	6	14	20

1.5 Delivery mode

- ☒ Face-to-face
- ☒ Online
- ☐ Intensive (e.g. residential, or summer school, weekend workshop - details below)

- ☐ Work-integrated learning activity
- ☒ Hybrid
- ☒ Full-time
- ☒ Part-time
- ☐ Other (please specify)

1.6 Pre-requisites and co-requisites

☒ Yes ☐ No

If **YES**, provide details of the prerequisite or co-requisite requirements below.

MFA501 - Mathematical Foundations of AI

1.7 Specialist facilities and/or equipment requirements

☒ Yes ☐ No

If **YES**, provide details of specialist facilities and/or equipment below.

Computer/laptop and relevant specialist software.

SECTION 2 – ACADEMIC DETAILS

2.1 Subject Descriptor

Deep learning is a subset of machine learning, and machine learning is a subset of AI.

This subject introduces the basics of Deep learning and how to build neural networks. Students will be presented with different methods and applications to different AI sub-areas such as natural language processing, speech recognition, and computer vision. The subject begins with the introduction of simple neural networks such as multi-layer perceptron, and progresses to complex concepts such as recurrent neural networks, convolutional networks, and long short-term networks.

2.2 Subject Learning Outcomes

Subject Learning Outcomes	Mapping to Course Learning Outcomes
a) Build, train and apply deep learning models to real-world tasks.	MSE(AI): CLO 3, CLO 6 MSE(AI, Adv): CLO 3, CLO 6
b) Compare and select ways to pre-process signals, images, and texts for natural language, speech recognition, and computer vision applications.	MSE(AI): CLO 3, CLO 8 MSE(AI, Adv): CLO 3, CLO 8
c) Develop critical analysis skills in deep learning research.	MSE(AI): CLO 3, CLO 7 MSE(AI, Adv): CLO 3, CLO 7
d) Demonstrate collaborative skills to apply deep learning to solve real world problems.	MSE(AI): CLO 1, CLO 6, CLO 7, CLO 8 MSE(AI, Adv): CLO 1, CLO 6, CLO 7, CLO 8
e) Demonstrate ability to effectively communicate scientific and technical information.	MSE(AI): CLO 5 MSE(AI, Adv): CLO 5

* The full list of Course Learning Outcomes and their descriptors is provided in Section 2.6 below.

2.3 Assessment Tasks

12-week duration

No.	Type and Description	Assessment due	Weighting	Subject Learning Outcome(s) assessed
1.	Title: Programming Problems Delivery Type: Individual Outcome: Summary Report and a source code Volume of assessment: 500 words (+/- 10%)	Module 4 (Week 4)	30%	a), b)
2.	Title: Deep Learning Project Proposal Presentation Delivery Type: Group Outcome: Report and a Presentation Volume of assessment: 1000 words(+/- 10%)	Module 8 (Week 8)	30%	b), c), d), e)
3.	Title: Deep Learning Final Project Delivery Type: Group Outcome: Report with a source code Volume of assessment: 1500 words(+/- 10%)	Wednesday, Module 12 (Week 12)	40%	c), d), e)

6-week duration

No.	Type and Description	Assessment due	Weighting	Subject Learning Outcome(s) assessed
1.	Title: Programming Problems Delivery Type: Individual Outcome: Summary Report and a source code Volume of assessment: 500 words (+/- 10%)	Module 4 (Week 2)	30%	a), b)
2.	Title: Deep Learning Project Proposal Presentation Delivery Type: Group Outcome: Report and a Presentation	Module 8 (Week 4)	30%	b), c), d), e)

	Volume of assessment: 1000 words(+/- 10%)			
3.	Title: Deep Learning Final Project Delivery Type: Group Outcome: Report with a source code Volume of assessment: 1500 words(+/- 10%)	Wednesday, Module 12 (Week 6)	40%	c), d), e)

2.4 Prescribed texts

These are core texts that are fundamental to meeting subject outcomes. Students may choose to purchase a copy. A limited number of these texts will be available for loan from the campus library.

N/A

2.5 Suggested readings and resources

These are supplementary readings and resources that support subject knowledge. These may be available from the campus library. Refer to *DLE602 Deep Learning* suggested readings and resources for the list of readings and resources.

Policies and Procedures, and related application forms, are available via the Torrens University Australia website.

2.6 Course Learning Outcomes

Course Learning Outcome Code	Course Title: Master of Software Engineering (Artificial Intelligence) Course Learning Outcome Descriptor
CLO 1	Analyse and integrate cognitive and technical skills from the software engineering body of knowledge to tackle complex commercial software engineering problems with innovation and ethical practice.
CLO 2	Critically analyse problem sets to synthesise acute software engineering solutions and communicate recommendations, conclusions and professional decisions to both expert and novice audiences.
CLO 3	Master and synthesise cognitive, technical and creative skills from the field of artificial intelligence necessary to effectively & efficiently solve or investigate problem sets.
CLO 4	Demonstrate sound judgement, non-partisanship and accountability as part of leadership role in a team to achieve effective planning, management, execution and communication.
CLO 5	Communicate technical solutions, recommendations and workarounds efficiently to a variety of audiences, including diverse multi-cultural groups.

CLO 6	Synthesise and abstract mathematical concepts, artificial intelligence fundamentals and industry best practices to solve varied commercial problems.
CLO 7	Possess a comprehensive and in-depth knowledge of Artificial Intelligence principles and techniques that includes understanding of recent developments.
CLO 8	Research complex problem sets to advance the field of artificial intelligence including those at the intersection of health, transportation, manufacturing or finance.

Course Learning Outcome Code	Course Title: Master of Software Engineering (Cloud Computing, Advanced) Course Learning Outcome Descriptor
CLO 1	Analyse and integrate cognitive and technical skills from the software engineering body of knowledge to tackle complex commercial software engineering problems with innovation and ethical practice.
CLO 2	Critically analyse problem sets to synthesise acute software engineering solutions and communicate recommendations, conclusions and professional decisions to both expert and novice audiences.
CLO 3	Master and synthesise cognitive, technical and creative skills from the field of artificial intelligence necessary to effectively & efficiently solve or investigate problem sets.
CLO 4	Demonstrate sound judgement, non-partisanship and accountability as part of leadership role in a team to achieve effective planning, management, execution and communication.
CLO 5	Communicate technically to present coherent and sustained arguments and to disseminate research that makes a contribution to knowledge.
CLO 6	Synthesise and abstract mathematical concepts, artificial intelligence fundamentals and industry best practices to solve varied commercial problems.
CLO 7	Possess a comprehensive and in-depth knowledge of Artificial Intelligence principles and techniques that includes understanding of recent developments.
CLO 8	Research complex problem sets to advance the field of artificial intelligence including those at the intersection of health, transportation, manufacturing or finance.